

COVID-19 Vaccination Rates and Death Rates

Title of Project (Research Question)

Is there a significant negative correlation between COVID-19 death rates and Vaccination rates over time?

Brief Project Idea Description

This project analyzes whether increases in COVID-19 vaccination rates are associated with decreases in COVID-19 death rates over time.

Project Overview

This project investigates whether there is a significant negative correlation between COVID-19 vaccination rates and COVID-19 death rates over time within a single country. This project applies exploratory data analysis and statistical inference techniques, including descriptive statistics, correlation analysis, hypothesis testing, and linear regression.

Dataset Description

The dataset consists of 100 dates ranging from 2021 to 2023. Each row contains observations for two key variables:

1. Vaccination Rate: The percentage of the population fully vaccinated on a given date.
 2. COVID-19 Death Rate: The number of deaths recorded on the same date.
- The data is derived from publicly available COVID-19 data repositories such as Our World in Data.